Shrishgovind Umesh Revankar

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SUMMARY

Embedded Systems Engineer with 2 years of EV industry experience in automotive electronics, CAN diagnostics. Skilled in embedded hardware-software integration, firmware development, communication protocols, and system-level debugging.

SKILLS

Programming & Software Development: Embedded C, C++, Python, .NET Framework, SystemVerilog.
Embedded Systems & Firmware: Bare-Metal Programming, .NET Desktop Application, Real-Time Operating System.
Communication Protocols: CAN (Controller Area Network), SPI, I2C, UART, Wi-Fi, Bluetooth.
Tools & Development Environment: Proteus, Keil uVision5, LTSpice, KiCAD, Vivado, Logic Analyzer, MATLAB, Cadence.
Hardware Skills: Wire & PCB Soldering, Signal Debugging, PCB Designing, Waveform Generator, EMI/EMC Testing.

PROJECTS

Self-Balancing Robot Control System (Graduate Research Project) — **May 2026**

- Developed a MATLAB-based control framework for a nonlinear two-wheeled self-balancing robot using state-space modeling, LMI-based stabilization, and data-driven control techniques.
- Designed and implemented model-based, data-driven, and dissipativity-based controllers using MATLAB, YALMIP, and semidefinite programming.
- Evaluated closed-loop stability, disturbance rejection, and transient performance through simulations of robot position, body angle, velocity, angular velocity, and control input.

Environmental Monitoring & Security System (Computing Principles for Mobile & Embedded Systems) — **May 2025**

- Configured low-level GPIO, I2C, SPI, and UART peripherals without HAL abstraction and developed bare-metal STM32U5 firmware for real-time sensor acquisition and communication.
- Transmitted real-time environmental data to a cloud platform via Wi-Fi for remote monitoring.
- Integrated matrix keypad authentication using GPIO row-column multiplexing and an I2C OLED display for access status feedback.

WORK EXPERIENCE

Tork Motors Pvt. Ltd. | Pune, Maharashtra, India — **June 2022 – May 2024**
Research & Development Engineer

- Architected and deployed a Renesas RH850-based HIL validation system for Vehicle Control Unit production testing, improving efficiency by 20% and adding encrypted traceability; prepared technical documentation and deployment guidelines for the VCU manufacturer, including on-site installation, live demonstration, and validation support.
- Developed production-cluster firmware for 3-Wheeler using Renesas RL78, to be configurable and achieving standardization of the code base and delivering a 10% improvement increase in overall system efficiency.
- Initiated the development of a proprietary software application using .NET Framework for automating CAN tests, streamlining the testing processes, improving the testing process by 30%, and with additional test cases.

CRCE Formula Racing – Formula Student | Mumbai, Maharashtra, India — **March 2019 – September 2021**
Electronics Head

- Led the team to develop a CAN-based telemetry system integrating internal and external sensors for real-time vehicle monitoring, enabling performance tuning and design optimization.
- Designed and prototyped a DC-motor-based semi-automatic transmission controller, applying LTSpice circuit simulation and hands-on lab debugging to validate motor driver behavior, power delivery, and vehicle-level integration this system helped in reducing the overall chassis packing and improved lap time by 1 second per lap.
- Designed and prototyped a safety-critical BSPD circuit using analog comparators and timer ICs, validating PCB-level behavior through LTSpice simulation and hands-on debugging to detect brake-throttle conflicts and trigger a 10-second shutdown.

EDUCATION

Stevens Institute of Technology | Hoboken, NJ, United States — **May 2026**
Master of Science, Computer Engineering **| GPA: 3.6 | Concentration:** Embedded Systems
Relevant Courses: Real-Time and Embedded Systems, Digital & Computer Systems Architecture, Applied Data Structure & Algorithms, Introduction to VLSI Design, Internet of Things, Autonomous Mobile Robotic System.

Fr. Conceicao Rodrigues College of Engineering | Mumbai, Maharashtra, India — **May 2022**
Bachelor of Engineering, Electronics Engineering